

Required Capstone Checkpoint 4.1: Advanced Retrieval Implementation

Directions

Download this worksheet and save a copy to your device before you begin working. Use this worksheet to plan, organize, and complete your submission for Required Capstone Checkpoint 4.1. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Completion Checklist

- ☐ Identification of a meaningful retrieval failure or limitation supported by evaluation evidence
- ☐ Analysis of the likely cause of the retrieval failure
- ☐ Selection and justification of an advanced retrieval strategy
- ☐ Evidence of implementation of the updated retrieval architecture
- ☐ Evaluation results comparing the updated system against the baseline system
- ☐ Reflection on the effectiveness of the architectural changes and any remaining limitations

Step 1. Retrieval Failure Identification

Identify a meaningful retrieval limitation in your current retrieval system.

- Describe the failure or limitation.
- Explain how the limitation affects retrieval performance.
- Reference evaluation evidence, test results, queries, or outputs that demonstrate the issue.

Retrieval failure or limitation
Response:

Evidence supporting the identified limitation
Response:

Step 2. Failure Analysis

Analyze why the retrieval failure occurs. Consider questions such as:

- Does the query require multiple retrieval steps?
- Is relevant information distributed across multiple documents, conversations, or entities?
- Is excessive irrelevant information being retrieved?
- Are important relationships within the data not represented effectively?
- Are there other factors contributing to the failure?

Likely cause(s) of the retrieval failure

Response:

Why the current retrieval architecture struggles

Response:

Step 3. Retrieval Strategy Selection

Select an advanced retrieval strategy designed to address the identified limitation. Examples may include:

- Query rewriting
- Query decomposition
- Retrieval fusion
- Multi-hop retrieval
- Relationship-aware retrieval

- Graph-based retrieval
- Another advanced retrieval approach

Selected retrieval strategy

Response:

Justification for the selected strategy

Response:

Step 4. Updated Retrieval Architecture

Describe the modifications made to your retrieval system.

- Summarize the architectural changes.
- Explain how the updated approach differs from the baseline system.
- Include implementation evidence where appropriate.

Description of architectural changes

Response:

Implementation evidence (e.g., screenshots, code snippets, retrieval outputs, notebook excerpts)

Response:

Step 5. Evaluation Results

Evaluate the updated system using the evaluation framework developed in previous modules.

- Present evaluation results.
- Compare performance against the baseline system.
- Include representative examples.

Evaluation Metric	Baseline Result	Updated Result

Example 1: Query, retrieved information, and result

Response:

Example 2: Query, retrieved information, and result

Response:

Example 3 (optional)

Response:

Step 6. Architectural Impact Analysis

Analyze the impact of the retrieval strategy.

- What improved?
- What remained challenging?
- Did the strategy address the original failure?

Improvements observed

Response:

Remaining limitations or challenges

Response:

Did the strategy address the original failure?

Response:

Step 7. Reflection and Next Steps

Reflect on the effectiveness of the architectural redesign.

- What did you learn from implementing this strategy?
- What would you improve next?
- Are there additional retrieval failures that still need to be addressed?

Reflection on the architectural redesign

Response:

Specific improvements or next steps

Response:

End of Required Capstone Checkpoint 4.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.